

Basin or Grounding? A Success-Conditioned Reach Readout for Vision–Language–Action Spatial Generalization

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Abstract

Vision–language–action (VLA) policies report steadily improving *spatial generalization*, almost always measured by task success rate. But a policy can succeed on a displaced object while never reaching toward it—executing a canonical, memorized motor program whose grasp basin is wide enough to occasionally succeed. Success rate cannot separate these cases. We introduce a cheap, interventional, displacement-normalized, *success-conditioned* reach readout: we displace the target object at episode start and measure where the pre-grasp, contact, and lift endpoints land along the axis from the object ($u=0$) to the canonical training location ($u=1$), split by success and failure. The readout asks a question success rate cannot: when a rollout *succeeds*, was the reach object-grounded or merely tolerated by a wide basin? Applying it to a competent SmolVLA across two LIBERO suites, we find that the same counterfactual-augmentation fine-tuning does opposite things by regime—in wide-basin LIBERO-spatial, successful reaches stay off the object (u_{succ} moves $1.06 \rightarrow 0.51$: success rises by widening the basin), while in tight-basin LIBERO-object, successful reaches become object-near (u_{succ} $0.67 \rightarrow 0.21$, displaced success $10 \rightarrow 27\%$). Success rate hides this inversion entirely. The readout is model-agnostic by construction and released as a drop-in evaluation tool.

1 Introduction

Vision–language–action (VLA) policies are increasingly evaluated on *spatial generalization*: can a policy trained on objects at certain locations still act when the object moves? The field’s answer is a number—task success rate under object displacement. That number is convenient, comparable, and, we argue, not diagnostic of what it is taken to mean.

Consider a policy that, when the object is displaced, reaches toward the *canonical* location where it saw the object during training, rather than toward the object’s new position. If the task’s grasp basin is wide—many pre-grasp poses lead to a successful grasp—such an anchored reach can still succeed a nontrivial fraction of the time. Success rate credits this as spatial generalization. A second policy that actually *grounds* to the displaced object and reaches for it produces the same success number. Success rate cannot tell the two apart, yet they differ in exactly the capability the metric claims to measure.

That VLA policies can anchor to training regularities is known, under names such as causal confusion [de Haan et al., 2019] and “memory traps” [Anonymous, 2025a]; that success-only evaluation is imperfect has also been argued [Anonymous, 2026a]. Our contribution is neither. It is a *readout* for a question these lines of work do not answer: on the rollouts that *succeed*, is the reach caused by the perceived object position (grounding), or is it anchored to the canonical location and merely tolerated by a wide basin?

We introduce an interventional, displacement-normalized, success-conditioned reach readout. We displace the target object by $\mathbf{d}$ at episode start and measure where the reach endpoint E lands, in the frame normalized by $\mathbf{d}$: $u = \langle \mathbf{e}, -\hat{\mathbf{d}} \rangle / \|\mathbf{d}\|$ runs from 0 at the object to 1 at the canonical location, and $\rho = \|\mathbf{e}\| / \|\mathbf{d}\|$ is the normalized error, with $\mathbf{e} = E - T$. We compute these at the pre-grasp, object-contact, and object-lift endpoints and, crucially, split them by success and failure. The readout is cheap (it reuses the rollout), model-agnostic (any policy that produces rollouts), and interpretable ($u \rightarrow 0$ means grounded, $u \rightarrow 1$ means anchored).

Using the readout on a competent SmolVLA [Anonymous, 2025c] across two LIBERO suites [Liu et al., 2023], we surface a finding that success rate hides. The *same* counterfactual-augmentation fine-tuning improves displaced success in both suites, but for opposite reasons (Figure 1). In wide-basin LIBERO-spatial, successful reaches stay *off the object* (u_{succ} moves from 1.06 to 0.51): success rises because the basin widens, not because the policy grounds. In

tight-basin LIBERO-object, successful reaches become *object-near* (u_{succ} moves from 0.67 to 0.21; displaced success rises from 10% to 27%): here the tight basin makes grounding necessary for success, so the same intervention is forced to ground. Success rate reports “improvement” in both and cannot distinguish the mechanisms.

Contributions.

1. A cheap, interventional, displacement-normalized, success-conditioned *reach readout* (u, ρ ; pre-grasp/contact/lift) that dissociates object-grounding from grasp-basin tolerance, released as a drop-in evaluation tool (Section 3).
2. The *basin-vs-grounding dissociation* on competent VLAs across two suites: displaced successes can be anchored, which success rate cannot reveal (Section 4).
3. The *cross-regime finding*: the same counterfactual-fine-tuning success gain reflects basin-widening in one suite and object-near reaching in another, invisible to success rate (Section 4).

A boundary analysis (Section 5) shows the readout also *rejects* tempting mechanistic explanations that do not survive across suites, illustrating its discriminating power. We scope our study to a single policy family and simulation; Section 6 states these limits and the model-agnostic path forward.

2 Related Work

Anchoring, shortcuts, and memory traps. That imitation and VLA policies latch onto training-distribution regularities rather than task-causal features is well documented. Causal confusion [de Haan et al., 2019] shows behavioral cloning exploits nuisance correlates of expert actions; object-aware regularization [Park et al., 2021] targets spurious object cues. In the VLA setting, “memory traps” [Anonymous, 2025a] name the failure of reproducing memorized trajectories instead of adapting to a moved scene, and propose an affordance-field fix; proprioceptive state has been identified as a shortcut that harms spatial generalization [Anonymous, 2025b]. We take the *existence* of anchoring as given. Our contribution is not the phenomenon but an instrument that measures, on the *successful* rollouts, whether the reach is grounded or anchored—a question these works do not pose.

Evaluating spatial generalization. Recent work argues that success-only benchmarks give incomplete or misleading pictures. LIBERO-X [Anonymous, 2026a] introduces a robustness litmus with progressive perturbations and a capability decomposition; “Robust Skills, Brittle Grounding” [Anonymous, 2026b] perturbs object locations and decomposes success into skill-execution versus instruction-grounding metrics. Spatial-tolerance and success-region analyses characterize *where* a policy works [Yang et al., 2025, Zhang et al., 2025, Anonymous, 2024]. These measure success degradation, capability axes, or spatial extent. None measures, on the rollouts that succeed, whether the endpoint moved with the displaced object or stayed at the canonical location—the basin-versus-grounding distinction (Table 1). Our readout is the manipulation-reach analogue of interventional acceptance tests for attribution [Adebayo et al., 2018]: it actively perturbs the cause and checks the effect.

Counterfactual data and fine-tuning geometry. We use counterfactual-augmentation fine-tuning purely as a controlled intervention, in the lineage of MimicGen [Mandlekar et al., 2023] and DemoGen [Xue et al., 2025]; we make no data-generation claim. Our boundary analysis (Section 5) connects to work on fine-tuning geometry—linear-probing versus full fine-tuning [Kumar et al., 2022], last-layer retraining for spurious correlations [Kirichenko et al., 2023], and robust weight interpolation [Wortsman et al., 2022]—which studies classification, not embodied reach.

3 The Reach Readout

Setup. We study SmolVLA [Anonymous, 2025c] fine-tuned to competence on the LIBERO-spatial and LIBERO-object suites [Liu et al., 2023] (10 tasks each), reaching 74% and 68% clean success respectively. All rollouts run in simulation with fresh observations at each planning step.

Table 1: What related evaluation work measures, versus the reach readout. Only the readout is conditioned on *successful* rollouts and separates basin-tolerance from grounding.

	Perturbs object?	Measures reach endpoint?	Conditioned on successes?	Separates basin from grounding?
Robust Skills, Brittle Grounding [Anonymous, 2026b]	yes	partial	no	no
LIBERO-X [Anonymous, 2026a]	yes	no	no	no
AFI / Memory Traps [Anonymous, 2025a]	yes	no	no	no
Reach readout (ours)	yes	yes	yes	yes

The reach readout

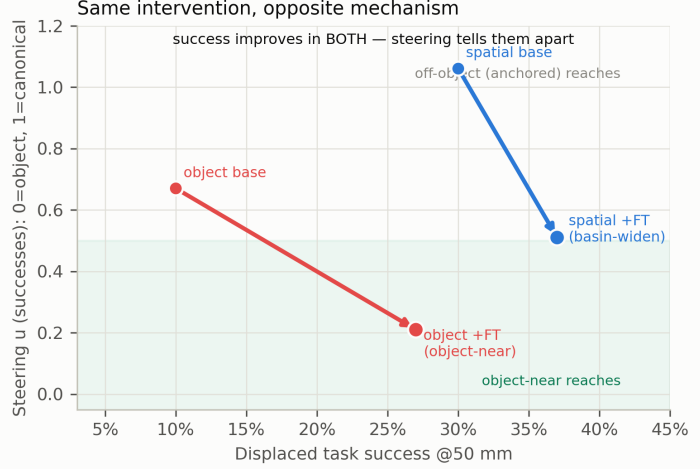
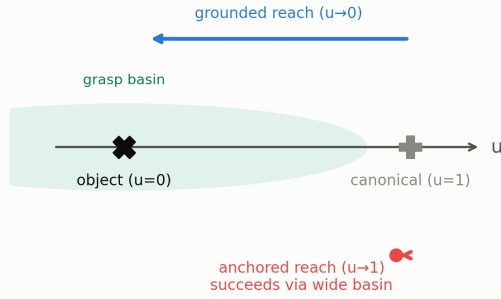


Figure 1: **The readout and the cross-regime inversion.** *Left:* we displace the object by $\mathbf{d}$ and measure the reach endpoint along u (object $u=0$, canonical $u=1$). A grounded reach lands near the object ($u \rightarrow 0$); an anchored reach lands near the canonical location ($u \rightarrow 1$) yet can still succeed inside a wide grasp basin. *Right (data):* the same counterfactual-fine-tuning intervention raises displaced success in both suites, but spatial successes stay off-object ($u_{\text{succ}} 1.06 \rightarrow 0.51$) while object successes become object-near ($u_{\text{succ}} 0.67 \rightarrow 0.21$). Success rate cannot distinguish the two.

The intervention. At episode start we displace the task-relevant object by a planar vector $\mathbf{d}$ (a grid of four magnitudes $\{25, 50, 75, 100\}$ mm $\times$ four directions), leaving the rest of the scene intact, then roll out the policy under its normal chunked execution. The canonical location C is the object’s training position; the displaced location is $T = C + \mathbf{d}$.

The readout. Let E be a reach endpoint and $\mathbf{e} = E - T$ its error relative to the displaced object. In the displacement-normalized frame we define

$$u = \frac{\langle \mathbf{e}, -\hat{\mathbf{d}} \rangle}{\|\mathbf{d}\|}, \quad v = \frac{\langle \mathbf{e}, \hat{\mathbf{d}}^\perp \rangle}{\|\mathbf{d}\|}, \quad \rho = \frac{\|\mathbf{e}\|}{\|\mathbf{d}\|}, \quad (1)$$

so $u = 0$ when the endpoint is at the object T and $u = 1$ when it is at the canonical location C ; v is the orthogonal component and ρ the normalized error magnitude. We extract three endpoints per rollout: *pre-grasp* (end-effector at the first gripper-close command), *contact* (first object motion), and *lift* (first object z -rise beyond a threshold). The pre-grasp endpoint captures the *planned* reach; contact and lift confirm whether the object was physically engaged.

Success conditioning. The defining move is to split u, v, ρ by rollout success. The question “do *successful* spatial-generalization rollouts ground to the object?” is exactly u_{succ} : if displaced successes have $u_{\text{succ}} \approx 1$, the policy succeeds while reaching the canonical location—basin tolerance, not grounding. We summarize a condition by the success-conditioned gap $\Delta u = \text{median}(u_{\text{fail}}) - \text{median}(u_{\text{succ}})$: $\Delta u > 0$ means successful reaches are pulled toward the object more than failing ones.

Pre-grasp endpoints in displacement-normalized frame — 50 mm; every cloud sits on canonical

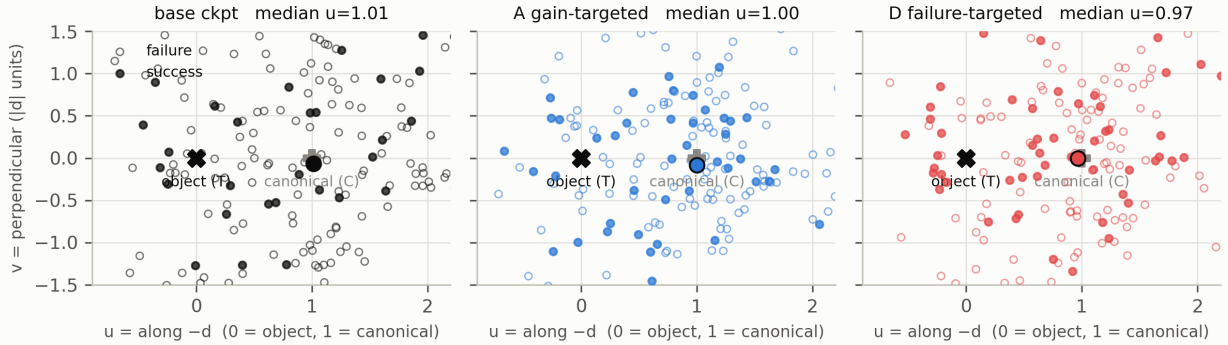


Figure 2: **Displacement-normalized pre-grasp endpoints**, success (filled) vs failure (open), for the base policy and fine-tuned conditions. Median u annotated. Successful reaches remain centered away from the object ($u \approx 1$) on the wide-basin suite; the readout exposes what success rate cannot.

Statistics. Endpoints are aggregated with a task-cluster (pairs) bootstrap over the 10 LIBERO tasks; we report 90% intervals. For “unchanged” claims we use two-one-sided-tests equivalence with pre-declared margins. Full statistical details, including a competence-matched degradation-null control used in Section 5, are in the appendix.

Reusability. The readout needs only object displacement at reset and endpoint logging; it is agnostic to the policy architecture and the task suite, and we release it as a drop-in tool.

4 Experiments

We ask three questions. Does the dissociation exist—can successful displaced rollouts be anchored (Section 4.1)? Is it invisible to success rate (Section 4.2)? And does the same intervention behave oppositely across regimes (Section 4.3)? Unless noted, displacement is 50 mm and u is the pre-grasp endpoint; $n = 200$ displaced rollouts per condition per magnitude.

4.1 The basin-grounding dissociation exists

On the base policies, displaced successes are anchored, not grounded. On LIBERO-object at 50 mm, the policy succeeds 10% of the time yet its successful pre-grasp reaches sit at $u_{\text{succ}} = 0.67$ —about two-thirds of the way to the canonical location, not at the object. On LIBERO-spatial the effect is stronger still: displaced successes sit near $u_{\text{succ}} = 1.06$. Success rate reports these as spatial-generalization successes; the readout shows the reach never grounded (Figure 2).

The contact and lift endpoints resolve an apparent tension—how can an anchored reach succeed? On LIBERO-spatial, successful rollouts have an anchored *plan* (pre-grasp $u = 0.51$ after fine-tuning) but engage the object late: the contact endpoint sits at $u = 0.22$ and lift at $u = 0.32$. The pre-grasp endpoint captures a planning bias toward the canonical location; success comes from late correction inside the basin, not from grounded planning. Measuring only success, or only the final endpoint, would miss this.

4.2 Success rate hides it

We fine-tune the base policy under five counterfactual-augmentation conditions that differ in where synthetic displaced demonstrations are placed. Across all five, the pre-grasp steering stays anchored—median u remains near 1.0 regardless of the condition’s displaced success—so success changes are *not* diagnostic of steering. A two-one-sided-tests equivalence check confirms the steering distributions are statistically indistinguishable across conditions within our

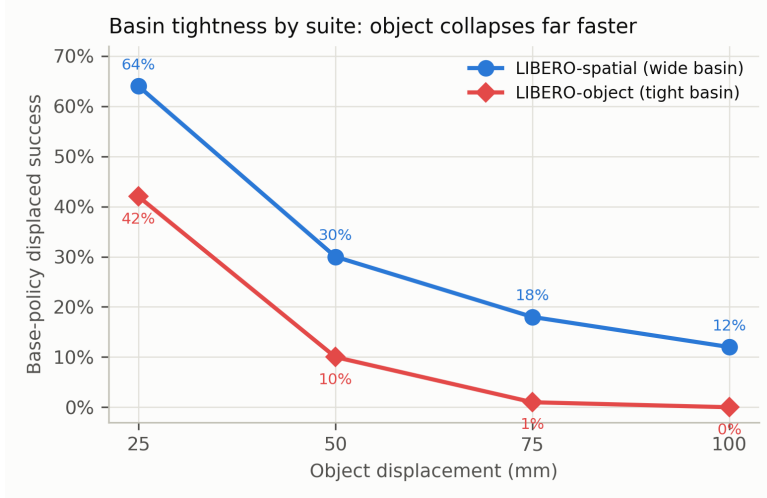


Figure 3: **Basin tightness by suite.** Base-policy displaced success versus displacement magnitude. LIBERO-object collapses far faster than LIBERO-spatial, so the same intervention that can basin-widen on spatial must ground on object.

Table 2: **Same intervention, opposite mechanism.** Displaced success and success-conditioned steering u_{succ} at 50 mm. Success rises in both suites; only the readout (u_{succ}) reveals that spatial stays off-object while object becomes object-near.

Suite	Condition	Clean	Displaced (50 mm)	u_{succ}
LIBERO-spatial	base	74%	31%	1.06
	+FT (M2)	52%	37%	0.51
LIBERO-object	base	68%	10%	0.67
	+FT (M2)	42%	27%	0.21

margins. A reader with access only to success rate would conclude that some conditions “generalize better”; the readout shows their reaches are equally anchored.

4.3 The cross-regime inversion

The central finding is that the same intervention—fine-tuning on 100% counterfactual (displaced) synthetic demonstrations, denoted M2—produces opposite mechanisms across suites (Table 2). On LIBERO-spatial, M2 raises displaced success from 31% to 37% but successful reaches remain off-object (u_{succ} moves only from 1.06 to 0.51): the gain is basin-widening. On LIBERO-object, M2 raises displaced success from 10% to 27% and successful reaches become object-near (u_{succ} moves from 0.67 to 0.21). The mechanism differs because the basin differs. Figure 3 shows the base policy’s displaced success collapsing far faster on object (42%→0% over 25→100 mm) than on spatial (64%→12%): object tasks have tight basins, so a policy cannot succeed on a displaced object *without* grounding, and the same intervention is forced to ground. Success rate reports “improvement” in both suites and is blind to the inversion.

5 Boundary Analysis: The Readout Rejects Tempting Mechanisms

A natural next step is to explain *why* the prior is anchored and to claim a fix. We report such an analysis not as a contribution but to show what the readout does when a tempting mechanistic story is put to it: it rejects the ones that do not hold up.

A defended-prior hypothesis. One hypothesis is that anchoring is defended by gradient competition with the canonical demonstrations. We test it by comparing M2 (100% synthetic) against N1 (the same synthetic count *plus* an equal count of canonical demonstrations, at matched training budget): if canonical demonstrations defend the prior, adding them back should re-anchor successful reaches. On LIBERO-spatial this direction appears— Δu drops from 0.43 (M2) to 0.23 (N1), a difference of +0.20 with a task-cluster 90% interval of $[-0.01, +0.46]$ (seed-level $[+0.16, +0.24]$). But on LIBERO-object the effect vanishes: +0.03, interval $[-0.15, +0.19]$. In tight object basins the canonical prior structurally fails—an anchored reach simply does not succeed—so it cannot defend via steering, and both M2 and N1 must ground. The hypothesis does not generalize.

Grounding beyond a competence-matched null. A second hypothesis is that fine-tuning produces grounding *beyond* what a merely degraded policy would show. We build a competence-matched degradation null—the base policy under inference-time action noise, tuned to match the intervention’s success—and ask whether the intervention’s success-conditioned steering exceeds the null’s. On LIBERO-spatial the direction is positive but underpowered: the pre-registered task-cluster statistic is +0.26 with interval $[-0.10, +0.59]$ (it excludes zero at the seed level, $[+0.15, +0.32]$). On LIBERO-object it is negative—the matched null grounds *more* than the intervention (null $\Delta u = 0.64$ vs 0.52), giving -0.12 , interval $[-0.40, +0.08]$: in a tight basin every surviving policy must ground, so the intervention adds nothing beyond degradation-survivorship.

Reading. Neither mechanistic claim survives across both suites. This is the point. The robust, general result is not a mechanism but the readout: it dissociates basin-widening from grounding, and it is sharp enough to reject explanations that success rate would leave unexamined. Full statistics—the degradation-null construction, the gap statistic, and the bootstrap procedure—are in the appendix.

6 Conclusion

We introduced a cheap, interventional, success-conditioned reach readout that asks a question success rate cannot: when a VLA policy appears to generalize spatially, including on the rollouts it gets right, is the reach grounded to the displaced object or anchored to the canonical training location and tolerated by a wide grasp basin? Using it on a competent SmolVLA across two LIBERO suites, we found that the same counterfactual fine-tuning basin-widens on a wide-basin suite and reaches object-near on a tight-basin suite—opposite mechanisms behind identical-looking success gains. A boundary analysis showed the readout also rejects tempting mechanistic explanations that fail to hold across suites, underscoring that the robust contribution is the instrument, not any single mechanism.

Limitations. Our study uses a single policy family (SmolVLA) and simulation (LIBERO). The readout is model-agnostic by construction—it needs only object displacement at reset and endpoint logging—but a second-policy datapoint is empirically pending: our attempts to obtain one (ACT and a pi0.5 checkpoint) were blocked by policy-integration and compute-contention issues, not by anything intrinsic to the readout. Seed budgets are set by a single-GPU regime; the mechanistic boundary results in Section 5 are correspondingly underpowered at the task-cluster level and are reported as such.

Future work. The readout is released as a drop-in tool; the immediate next step is to apply it to additional policy families and to real-robot rollouts, where basin geometry and grounding may diverge further. Contact- and lift-time endpoints suggest a finer question—when and where an anchored plan is corrected—that the readout is already positioned to measure.

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A Statistical Details

Task-cluster bootstrap. LIBERO suites contain 10 tasks, which we treat as clusters. All intervals are 90% percentile intervals from a nonparametric pairs-cluster bootstrap: we resample the 10 task clusters with replacement and recompute the median statistic. (This is not the Rademacher wild-cluster bootstrap, which is defined for regression coefficients rather than medians; a beta-binomial / logistic mixed model with task random effects is the appropriate pre-registered inference for a camera-ready version.)

Success-conditioned gap. For a condition, $\Delta u = \text{median}(u_{\text{fail}}) - \text{median}(u_{\text{succ}})$ over displaced rollouts at 50 mm, with u the pre-grasp endpoint (closest-approach fallback when the gripper never closes; fallback was not triggered in the reported runs).

Competence-matched degradation null. The null is the base policy under inference-time Gaussian action noise added to the executed end-effector deltas. We search the noise scale so that the null’s clean and displaced success match the intervention arm; the matched scales are $\sigma \approx 0.15$ (spatial) and $\sigma \approx 0.20$ (object). The null shares the base policy’s grounding and is degraded only by execution noise, isolating whether the intervention grounds *beyond* degradation-survivorship. We report the directional gain $G = \Delta u_{\text{int}} - \Delta u_{\text{null}}$ with a v -dispersion guard requiring the intervention not to be more dispersed than the null.

Seeds. Claim-critical fine-tuning arms are run at 3 training seeds; the reported spatial Δu for the 100%-synthetic arm is 0.43 ± 0.02 across seeds, indicating the steering signal is not seed noise.

B Additional Figures

Supplementary figures—success-by-condition bars, the success-versus-steering scatter, realized synthetic-displacement distributions, and the full dose–response of steering versus synthetic fraction—are provided in the released artifact.

C Second-Policy Integration Notes

We attempted a second-policy datapoint with a from-scratch ACT policy and a pretrained pi0.5 LIBERO checkpoint. ACT trained (training loss decreased) but evaluated at near-zero clean success in our rollout harness, consistent with an evaluation-interface mismatch rather than a training failure; the pi0.5 attempt was repeatedly interrupted by checkpoint-download and shared-GPU contention. We report these as engineering limitations; the readout itself is policy-agnostic and imposes no such requirement.